Global School ERP: application flow and summary

Oct 14, 2026 · @abhi

Summary

Global School ERP runs a school group's whole academic year in one system: admissions, students, staff, classes, timetable, attendance, exams, fees, payroll, parent communication and year-end rollover.

One account can hold several schools in different countries, each with its own currency, time zone, language and curriculum. The demo shows a CBSE school in Bengaluru (INR, English) and an IB school in Dubai (AED, Arabic, right-to-left).

The current version is a working first release: every screen described here runs end to end, and 178 automated checks pass. It is built on ASP.NET Core (.NET 10) and Angular 20.

Who uses it

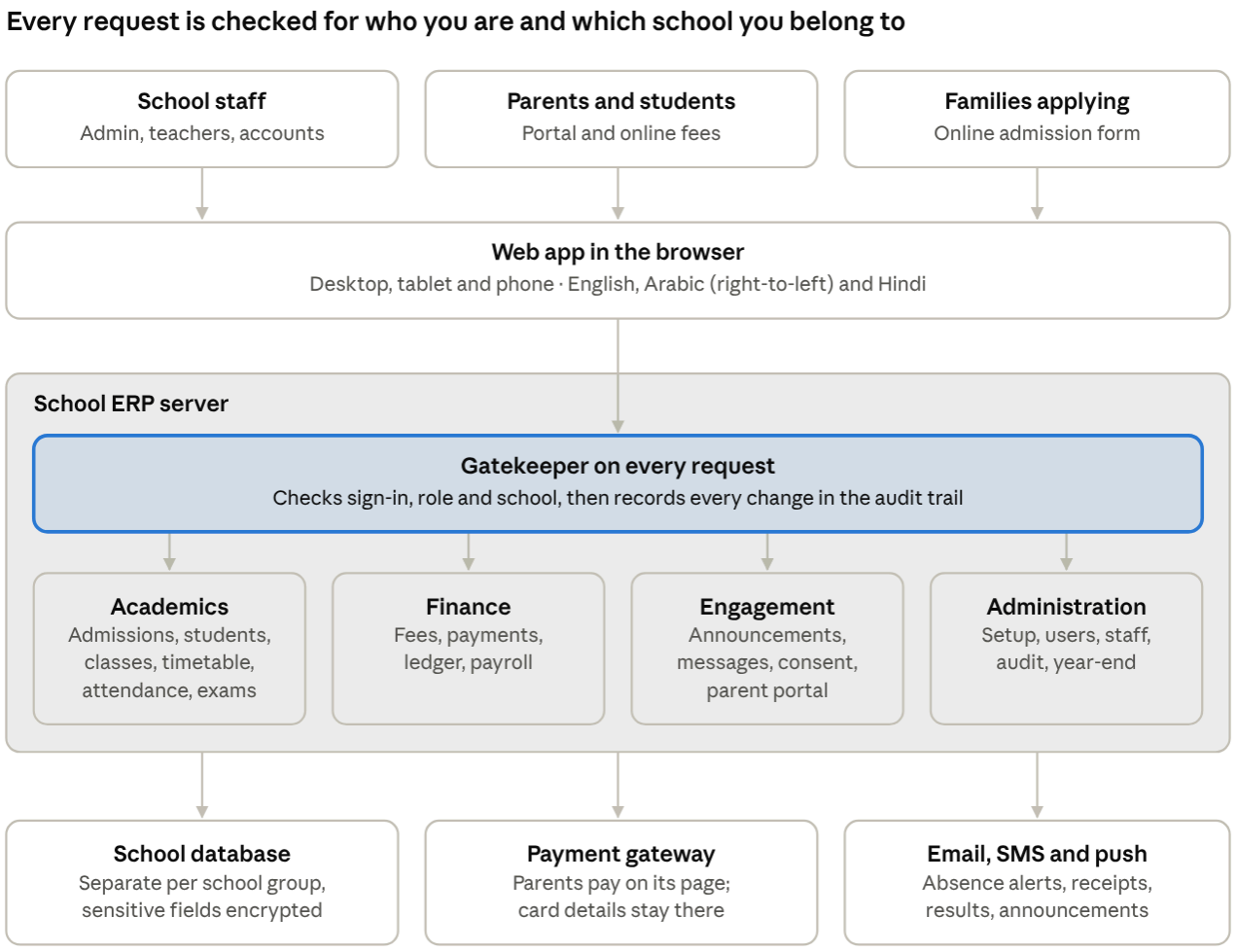
Each person signs in and sees only the screens and records their role allows. Schools can copy and edit these roles.

Role	Sees	Main work
School admin	The whole school group	Setup, users and roles, imports, every module
Principal	Own campus	Approve and publish results, leave approvals, announcements, consent forms
Teacher	Own classes and subjects	Mark attendance, enter marks, message parents, own payslips
Accountant	Own campus finance	Fee structures, invoices, payments, ledger, payroll
Parent	Own children only	Attendance, timetable, published results, pay fees, give consent, message teachers
Student	Self	Timetable and published results
Platform support	Tenant directory only	Read-only access to a school's data, and only after that school approves it

Staff sign in with email and password, with optional two-step codes from an authenticator app. Parents sign in with a one-time code sent to their email or phone.

How the system fits together

Everyone uses the same web app; the server checks each request before any module touches data.

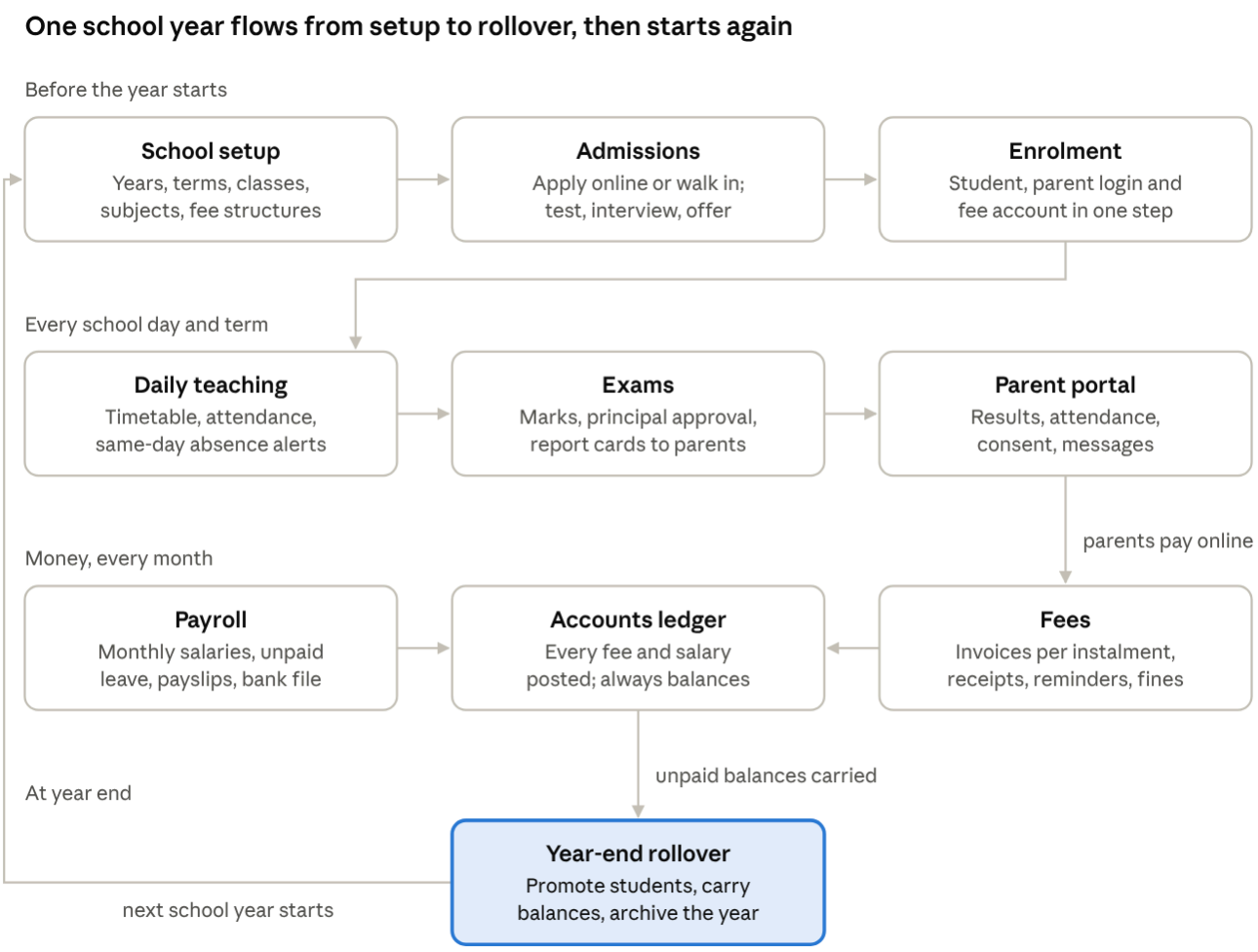


system overview · users, web app, server, connected services

The gatekeeper decides what each person may see, so a teacher, a parent and an accountant using the same screen see different records.

The school year, end to end

A year starts with setup and admissions, runs on daily teaching and monthly money, and ends with a rollover that opens the next year.



school year flow · 10 stages, one loop

Fees and salaries both post to one accounts ledger, so the school's books stay in balance all year. The rollover promotes every student, copies the year's setup and carries unpaid fees forward.

Key journeys

Four journeys carry most of the school's daily work. Each one is a few clicks, and every step is recorded in the audit log.

Applicant to enrolled student

1. A family fills in the school's public application form and uploads documents, or the front office records a walk-in.
2. The application moves across the pipeline: enquiry, applied, assessed, offered, waitlisted or rejected. Tests and interviews are scheduled and the family is notified.
3. Enrol places the child in a class. One step creates the student record, the parent's portal login, the family fee account and the class enrolment.

4. The first invoices are raised from the class's fee structure.

Marks to report card

1. The exams office sets up an assessment plan per subject: components such as class test and term exam, their weights, and the grading scheme (CBSE, IB, Cambridge, GPA or the school's own).
2. The subject teacher enters marks before the deadline and submits them. Submitted marks are locked.
3. The principal approves, or reopens for correction.
4. Publishing releases results to parents, who are notified. Report cards print as PDF in English, Arabic (right-to-left) or Hindi.

Invoice to payment

1. The accountant generates invoices for the year. Each student gets one per instalment, with scholarships and sibling discounts applied.
2. Parents pay online from the portal through the payment gateway, or the office records cash, cheque or bank payments.
3. A receipt is issued and the parent is notified. Overdue invoices attract late fines and reminders.
4. Every invoice, payment, refund, credit note and fine is posted to a double-entry ledger that always balances.

Monthly payroll

1. Each staff member has a salary structure: Basic, allowances, and deductions such as provident fund and professional tax.
2. Run payroll for a month creates draft payslips and deducts any approved unpaid leave.
3. Approval books the salary cost to the ledger and releases payslips to staff.
4. Mark as paid records the bank payment. The accountant downloads a bank transfer file and payslip PDFs.

Modules

Fourteen modules, each a menu item, share one set of student, staff and class records.

Module	What it does
Dashboard	Headline numbers, attendance trend, fee collection, class sizes, admissions pipeline

Module	What it does
School setup	Campuses, academic years and terms, grades, subjects, periods, attendance codes, custom fields
Admissions	Public application form, pipeline board, tests and interviews, one-step enrolment
Students	Profiles, guardians and siblings, documents, medical notes (restricted), transfers, transfer certificates
Staff	Teacher profiles, subjects and classes taught, weekly timetable, leave, salary structure
Classes	Class teacher, students, subjects and teachers, attendance per student, results status
Timetable	Subject and teacher mapping, weekly grid, teacher and room clash checks
Attendance	Daily or per-period registers, absence alerts to parents, summaries, staff leave requests
Exams	Assessment plans, grading schemes, marks entry, approval, publishing, report card PDFs
Fees	Fee structures, instalments, discounts, invoices, online and office payments, refunds, late fines, aging report, ledger
Payroll	Monthly payroll runs, unpaid leave deductions, payslip PDFs, bank transfer file
Communication	Announcements by campus, grade or class; consent forms; delivery log; parent and teacher messages
Parent portal	Children's attendance, timetable, results, fees and payment, consent
Security and year end	Audit log, data export and erasure, support access approvals, two-step sign-in, year-end rollover

Security, privacy and multi-school support

Each school group's data is walled off from every other customer. An automated test confirms that no screen can read another group's records.

- **Separate data per customer.** Every record carries its school group, and every lookup is filtered by the signed-in user's group.
- **Need-to-know access.** Teachers see only their own classes, parents only their own children. Medical notes, ID numbers, bank accounts and tax IDs are encrypted and shown only to authorised roles.
- **Full audit trail.** Every create, change, delete, export and sign-in is recorded with who, when and what changed. The log cannot be edited from the application.
- **School-controlled privacy.** The school can export everything held about a student, or erase a former student's personal data. Financial records are kept but no longer identify the child.
- **Support access needs approval.** The software provider's staff can only view a school's data after the school approves a request. Access is read-only, time-limited and revocable at once.
- **Payments.** Card details are entered on the payment provider's page and never reach the school's servers.
- **Many languages and countries.** English, Arabic (right-to-left) and Hindi interface; money, dates and time zones follow each campus.

Built now, and what comes next

Everything in the module table works today against demo data. The items below are what stands between this version and a live pilot school.

Area	Today	Next step
Online payments	Demo payment page	Connect Razorpay, Stripe and one Gulf gateway
Email, SMS, push	Recorded in the system log; in-app notices work	Connect an email service and an SMS provider per region
Database	SQLite for the demo	Move to PostgreSQL or SQL Server with managed migrations
Sign-in	Two-step codes optional	Make two-step required for staff; add Google and Microsoft sign-in
Mobile	Web app works on phones and tablets	Installable parent app with offline attendance
Payroll	Salary structures, runs, payslips, bank file	Country packs: India PF, ESI and TDS; Saudi GOSI; UAE wage protection file

Area	Today	Next step
Translations	English complete; Arabic and Hindi for menus, dashboard and parent portal	Full Arabic and Hindi for every staff screen
Hosting	Runs on a developer machine	Cloud hosting per region, backups, load and penetration tests

Later phases from the plan: homework and LMS, transport with GPS, HR recruitment and appraisals, reports builder, hostel, library and inventory.